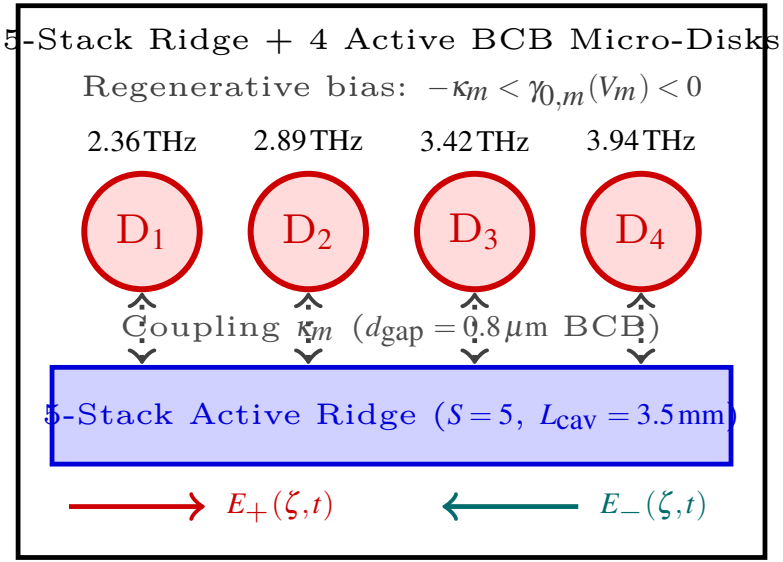
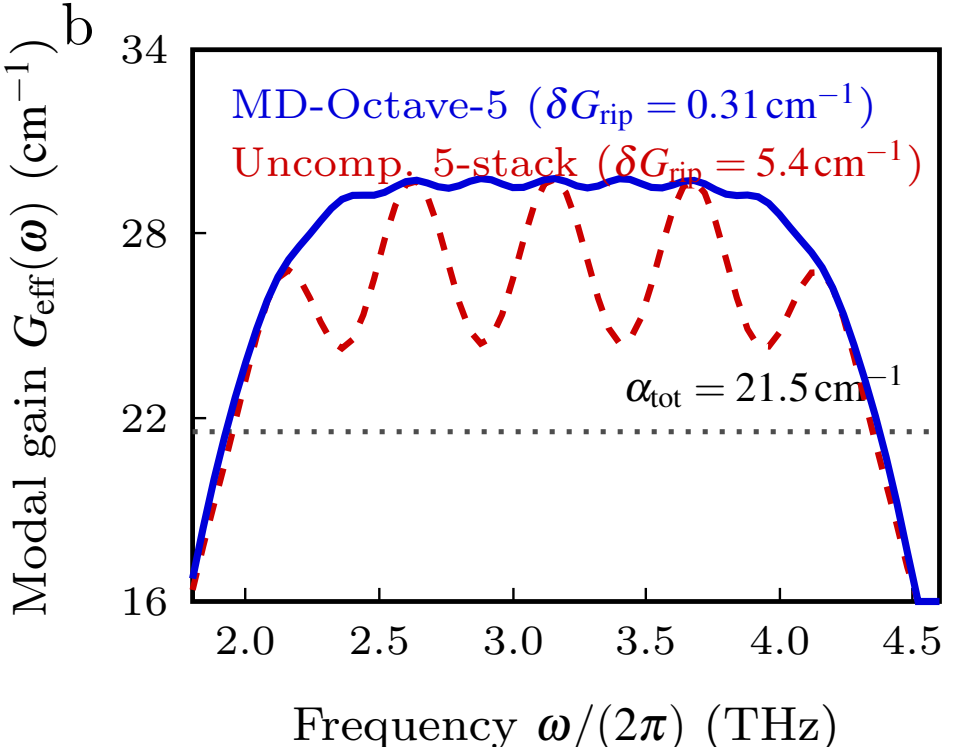
a


5-Stack Ridge + 4 Active BCB Micro-Disks
Regenerative bias: $-\kappa_m < \gamma_{0,m}(V_m) < 0$
2.36 THz
2.89 THz
3.42 THz
3.94 THz
D1
D2
D3
D4
Coupling κ_m ($d_{\rm gap} = 0.8\,\mu$m BCB)
5-Stack Active Ridge ($S = 5$, $L_{\rm cav} = 3.5$ mm)
$E_+(\zeta,t)$
$E_-(\zeta,t)$

b


MD-Octave-5 ($\delta G_{\rm rip} = 0.31\,{\rm cm}^{-1}$)
Uncomp. 5-stack ($\delta G_{\rm rip} = 5.4\,{\rm cm}^{-1}$)
$\alpha_{\rm tot} = 21.5\,{\rm cm}^{-1}$
Modal gain $G_{\rm eff}(\omega)$ (cm^{-1})
Frequency $\omega/(2\pi)$ (THz)
34
28
22
16
2.0
2.5
3.0
3.5
4.0
4.5